AMC12/COMC Strategic Problem Analysis
2016 AMC 12A Problem 15

Competition Math Framework

October 3, 2025

Problem Statement

Circles with centers P, Q and R, having radii $1, 2$ and 3, respectively, lie on the same side of line l and are tangent to l at P', Q' and R', respectively, with Q' between P' and R'. The circle with center Q is externally tangent to each of the other two circles. What is the area of triangle PQR?

(A) 0 **(B)** $\sqrt{\frac{2}{3}}$ **(C)** 1 **(D)** $\sqrt{6}-\sqrt{2}$ **(E)** $\sqrt{\frac{3}{2}}$

Problem Configuration Analysis

A. Problem Classification

- **Problem Type**: To Find - calculate the area of triangle PQR
- **Difficulty Band**: Mid-band (Problem 15/25, transitioning to late-band)
- **Competition Context**: AMC12 (75 min, 25 problems) - Target time 4-5 minutes
- **Mathematical Domain**: Plane Geometry with strong algebraic computation component
- **Source Context**: 2016 AMC 12A Problem 15 - historically correct answer rate 15-20%

B. Problem Structure Identification

- **Given Information**:
  - Three circles with radii 1, 2, 3 centered at P, Q, R respectively
  - All circles tangent to line l from same side
  - Tangent points P', Q', R' with Q' between P' and R'
  - Circle Q externally tangent to both circles P and R
- **Constraints**:
  - External tangency implies $PQ = 1 + 2 = 3$ and $QR = 2 + 3 = 5$
  - Tangency to line implies heights are the radii (1, 2, 3)
  - Ordering constraint: Q' between P' and R'
- **Objective**: Find area of triangle PQR from answer choices
- **Hidden Assumptions**:
  - Configuration is uniquely determined by constraints
  - Triangle exists (non-collinear points)

  - Answer involves nested radicals (based on choices)

- **Answer Choice Leverage**:
  - (A) is trivial case - immediately eliminable (tangency conditions imply non-degeneracy)
  - (B), (C), (E) are relatively simple expressions
  - (D) $\sqrt{6} - \sqrt{2}$ is most complex - likely answer if problem is difficult
  - Can verify by checking if calculated area matches a choice

C. Strategic Orientation

- **Hypothesis**: The tangency conditions and external tangency determine unique configuration
- **Conclusion**: Area can be computed via coordinate geometry or trapezoid decomposition
- **Notation**:
  - Centers: P, Q, R
  - Radii: $r_P = 1$, $r_Q = 2$, $r_R = 3$
  - Tangent points: P', Q', R' on line l
  - Distances: $PQ = 3$, $QR = 5$, $PR =?$
- **Intuitive Approach**: Use trapezoid decomposition method - break into manageable pieces
- **Cross-Domain Potential**:
  - Coordinate geometry: Place line l as x-axis
  - Pythagorean theorem: Find horizontal distances between tangent points
  - Heron's formula: After finding all three side lengths

Strategic Framework Application

A. Psychological Strategies

- **Mental Toughness**: This is a mid-band problem requiring careful calculation. Multiple approaches exist - if one gets messy, switch methods.
- **Creativity Cultivation**: The trapezoid decomposition approach is elegant - visualize breaking the configuration into trapezoids and triangles.
- **Iterative Refinement**: If initial calculation produces messy algebra, simplify by choosing better coordinate system or decomposition strategy.

B. Investigation Strategies

- **Startup Orientation**: Draw accurate diagram showing three circles tangent to line with proper ordering
- **Get Your Hands Dirty**: Calculate $P'Q'$ and $Q'R'$ using Pythagorean theorem on trapezoids
- **Penultimate Step**: After finding all trapezoid areas, final step is simple addition/subtraction
- **Wishful Thinking**: "If only we knew PR..." - but we can find it via Pythagorean theorem
- **Make It Easier**: Instead of finding PR directly, use trapezoid area method to avoid that calculation

- **Conjecture via Small Cases**: Not applicable - unique configuration
- **Visualization**: Critical - draw perpendiculars from centers to line to form trapezoids

C. Late-Band Strategic Playbook

- **Answer-Choice Leverage**: Answer (D) $\sqrt{6} - \sqrt{2}$ suggests problem involves nested radicals - consistent with Pythagorean calculations yielding $\sqrt{8}$ and $\sqrt{24}$
- **Make It Easier**: Not used - problem structure is already clean
- **Penultimate Step**: Final calculation is adding trapezoid areas and subtracting large trapezoid
- **Wishful Thinking**: Not needed - direct approach works
- **Conjecture via Small Cases**: Verify answer makes sense: area should be positive and reasonable given side lengths 3, 5

D. Argument Construction Strategy

- **Direct Proof**: Yes - calculate areas using trapezoid decomposition method
- **Proof by Contradiction**: Not applicable
- **Mathematical Induction**: Not applicable
- **Stylistic Conventions**: WLOG can assume coordinate system with l as x-axis

E. Cross-Domain Translation

- **Draw a Picture**: Essential - diagram reveals trapezoid structure
- **Recast the Problem**: "Find area" becomes "decompose into trapezoids and calculate"
- **Change Your Point of View**: Coordinate geometry approach: Shoelace formula after finding coordinates

Tactical Methodology Selection

A. Core Mathematical Tactics

- **Symmetry**: Configuration has no obvious symmetry, but trapezoid method exploits structural regularity
- **The Extreme Principle**: Not applicable
- **The Pigeonhole Principle**: Not applicable
- **Invariants**: The heights (radii) remain constant throughout calculation

B. Crossover Tactics

- **Graph Theory**: Not applicable
- **Complex Numbers**: Not applicable (though could represent as complex plane)
- **Generating Functions**: Not applicable
- **Coordinate Geometry**: Highly applicable - primary alternative approach using Shoelace theorem

Conversion Techniques Application

Recognition Index Analysis

Primary Recognition Triggers (1-3 seconds):

1. **Circles tangent to line** → Heights equal radii (perpendicular from center to tangent point)
2. **External tangency** → Distance between centers equals sum of radii
3. **Find area of triangle** → Multiple approaches: coordinate geometry, Heron's, decomposition
4. **Answer choices with nested radicals** → Expect Pythagorean theorem calculations

A. High-Probability Techniques Selected

1. **Coordinate Geometry Conversion**:
   - Place P' at origin, line l as x-axis
   - Centers: $P = (0, 1)$, $Q = (P'Q', 2)$, $R = (P'Q' + Q'R', 3)$
   - Find $P'Q'$ and $Q'R'$ using Pythagorean theorem
   - Apply Shoelace formula for area
2. **Pythagorean Theorem Application**:
   - In trapezoid $PP'Q'Q$: $P'Q'^2 + (r_Q - r_P)^2 = PQ^2$
   - $P'Q'^2 + 1^2 = 3^2 \Rightarrow P'Q' = \sqrt{8} = 2\sqrt{2}$
   - In trapezoid $QQ'R'R$: $Q'R'^2 + (r_R - r_Q)^2 = QR^2$
   - $Q'R'^2 + 1^2 = 5^2 \Rightarrow Q'R' = \sqrt{24} = 2\sqrt{6}$
3. **Trapezoid Area Decomposition**:
   - Area of trapezoid with parallel sides a, b and height h: $\frac{(a+b)h}{2}$
   - $[PP'Q'Q] = \frac{(r_P+r_Q)\cdot P'Q'}{2} = \frac{(1+2)\cdot 2\sqrt{2}}{2} = 3\sqrt{2}$
   - $[QQ'R'R] = \frac{(r_Q+r_R)\cdot Q'R'}{2} = \frac{(2+3)\cdot 2\sqrt{6}}{2} = 5\sqrt{6}$
   - $[PP'R'R] = \frac{(r_P+r_R)\cdot P'R'}{2} = \frac{(1+3)(2\sqrt{2}+2\sqrt{6})}{2} = 4\sqrt{2} + 4\sqrt{6}$
4. **Area Addition/Subtraction**:
   - Key equation: $[PP'Q'Q] + [QQ'R'R] = [PQR] + [PP'R'R]$
   - Therefore: $[PQR] = [PP'Q'Q] + [QQ'R'R] - [PP'R'R]$
   - $[PQR] = 3\sqrt{2} + 5\sqrt{6} - (4\sqrt{2} + 4\sqrt{6})$
   - $[PQR] = 3\sqrt{2} + 5\sqrt{6} - 4\sqrt{2} - 4\sqrt{6}$
   - $[PQR] = -\sqrt{2} + \sqrt{6} = \sqrt{6} - \sqrt{2}$

B. Medium-Probability Techniques

1. **Heron's Formula**:
   - Would require finding PR first via Pythagorean theorem
   - $PR^2 = (P'R')^2 + (r_R - r_P)^2 = (2\sqrt{2} + 2\sqrt{6})^2 + 4$
   - Leads to nested radical: $PR = 2\sqrt{9 + 4\sqrt{3}}$
   - More complex but verifiable approach

C. Technique Selection Rationale

- **Primary Method**: Trapezoid decomposition - clean, direct, avoids finding PR
- **Alternative Method**: Coordinate geometry with Shoelace - equally clean
- **Backup Method**: Heron's formula - more computation but verifies answer
- **Why Chosen**: Trapezoid method is fastest and least error-prone for competition

D. Integration Strategy

- All three methods should yield $\sqrt{6} - \sqrt{2}$
- Trapezoid method integrates: Pythagorean theorem + area formula + algebraic manipulation
- No conflicts between techniques - they complement each other

Verification Strategy Design

A. Verification Level Selection

Based on decision tree for mid-band problem with computational complexity:
Selected Level: Level 4 - Standard Verification

- Difficulty: Mid-band (P15)
- Computation: Moderate (multiple Pythagorean calculations)
- Confidence: Should be high after clean trapezoid calculation
- Time available: 1-1.5 minutes for verification

B. Specific Verification Methods

1. **Dimensional Analysis**:
   - Area should have dimension $[\text{length}]^2$
   - $\sqrt{6} - \sqrt{2}$ is dimensionless number, multiply by appropriate unit squared
   - Check: Expression is positive ($\sqrt{6} \approx 2.45 > \sqrt{2} \approx 1.41$) ✓
2. **Reasonableness Check**:
   - Triangle has sides $PQ = 3$, $QR = 5$
   - Rough upper bound: $\frac{1}{2} \cdot 3 \cdot 5 = 7.5$
   - Computed: $\sqrt{6} - \sqrt{2} \approx 2.45 - 1.41 = 1.04$
   - Reasonable - triangle is not very "tall" ✓
3. **Answer Choice Matching**:
   - Computed answer exactly matches choice (D)
   - No other choice is close numerically ✓
4. **Algebraic Verification**:
   - Re-check: $3\sqrt{2} + 5\sqrt{6} - 4\sqrt{2} - 4\sqrt{6}$
   - Group terms: $(3-4)\sqrt{2} + (5-4)\sqrt{6} = -\sqrt{2} + \sqrt{6}$ ✓

C. Specialized Verification

Geometry-Specific Checks:

- Verify $P'Q' = 2\sqrt{2}$: $\sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}$ ✓
- Verify $Q'R' = 2\sqrt{6}$: $\sqrt{5^2 - 1^2} = \sqrt{24} = 2\sqrt{6}$ ✓
- Triangle inequality: $3 + 5 > PR$, $|5 - 3| < PR$ (should be satisfied)

D. Confidence Calibration

- **Initial Confidence**: 85% (clean calculation, matches answer choice)
- **After Verification**: 95% (all checks pass, answer is exact match)
- **Flag for Review**: No - high confidence, move to next problem

Solution Roadmap

Complete Solution Roadmap

Time Estimate: 4-5 minutes

Step 1: Setup and Diagram (30 seconds)

- Draw line l horizontally
- Draw three circles tangent to l from above
- Mark radii: 1, 2, 3 for circles P, Q, R
- Note ordering: P', Q', R' left to right
- Mark tangency: Q externally tangent to P and R

Checkpoint: Diagram should clearly show configuration, verify $PQ = 3$, $QR = 5$

Step 2: Find Horizontal Distances (1.5 minutes)

- For trapezoid $PP'Q'Q$:
  - Vertical difference: $|r_Q - r_P| = |2 - 1| = 1$
  - Hypotenuse: $PQ = 3$
  - Horizontal: $P'Q' = \sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}$
- For trapezoid $QQ'R'R$:
  - Vertical difference: $|r_R - r_Q| = |3 - 2| = 1$
  - Hypotenuse: $QR = 5$
  - Horizontal: $Q'R' = \sqrt{5^2 - 1^2} = \sqrt{24} = 2\sqrt{6}$

Checkpoint: $P'Q' = 2\sqrt{2}$, $Q'R' = 2\sqrt{6}$, therefore $P'R' = 2\sqrt{2} + 2\sqrt{6}$

Step 3: Calculate Trapezoid Areas (1.5 minutes)

- Area of $PP'Q'Q$:

$$[\text{PP'Q'Q}] = \frac{(r_P + r_Q) \cdot P'Q'}{2} = \frac{(1+2)(2\sqrt{2})}{2} = 3\sqrt{2}$$

- Area of $QQ'R'R$:

$$[\text{QQ'R'R}] = \frac{(r_Q + r_R) \cdot Q'R'}{2} = \frac{(2+3)(2\sqrt{6})}{2} = 5\sqrt{6}$$

- Area of $PP'R'R$:

$$\begin{aligned}[\text{PP'R'R}] &= \frac{(r_P + r_R) \cdot P'R'}{2} \\ &= \frac{(1+3)(2\sqrt{2} + 2\sqrt{6})}{2} \\ &= \frac{4(2\sqrt{2} + 2\sqrt{6})}{2} \\ &= 2(2\sqrt{2} + 2\sqrt{6}) \\ &= 4\sqrt{2} + 4\sqrt{6}\end{aligned}$$

Checkpoint: Three trapezoid areas calculated, ready for final combination

Step 4: Apply Area Equation (1 minute)

Key relationship: $[PP'Q'Q] + [QQ'R'R] = [PQR] + [PP'R'R]$
Therefore:

$$\begin{aligned}[PQR] &= [PP'Q'Q] + [QQ'R'R] - [PP'R'R] \\ &= 3\sqrt{2} + 5\sqrt{6} - (4\sqrt{2} + 4\sqrt{6}) \\ &= 3\sqrt{2} + 5\sqrt{6} - 4\sqrt{2} - 4\sqrt{6} \\ &= (3-4)\sqrt{2} + (5-4)\sqrt{6} \\ &= -\sqrt{2} + \sqrt{6} \\ &= \sqrt{6} - \sqrt{2}\end{aligned}$$

Checkpoint: Answer matches choice (D)

Step 5: Verification (30 seconds)

- Check sign: $\sqrt{6} > \sqrt{2}$, so area is positive ✓
- Numerical check: $\sqrt{6} - \sqrt{2} \approx 2.45 - 1.41 = 1.04$ ✓
- Answer choice match: (D) ✓

Alternative Approaches

1. **Coordinate Geometry + Shoelace**:
   - Place $P = (0, 1)$, $Q = (2\sqrt{2}, 2)$, $R = (2\sqrt{2} + 2\sqrt{6}, 3)$
   - Apply Shoelace formula
   - Time: Similar to trapezoid method

2. **Heron's Formula**:
   - Find $PR = 2\sqrt{9 + 4\sqrt{3}}$
   - Apply Heron's with sides 3, 5, $2\sqrt{9 + 4\sqrt{3}}$
   - Time: 6-7 minutes (more complex)

Pitfall Guide

1. **Arithmetic Error**: Miscalculating $\sqrt{8}$ or $\sqrt{24}$
   - Prevention: Double-check simplification of radicals
   - Recovery: Verify with calculator if allowed
2. **Area Equation Error**: Wrong combination of trapezoid areas
   - Prevention: Draw diagram clearly showing which areas add/subtract
   - Recovery: Re-derive from first principles
3. **Sign Error**: Getting $\sqrt{2} - \sqrt{6}$ (negative area)
   - Prevention: Always check final answer is positive
   - Recovery: Review area combination equation
4. **Pythagorean Theorem Application**: Using wrong sides
   - Prevention: Label trapezoid clearly with known quantities
   - Recovery: Verify vertical difference is radius difference

Time Management Checkpoints

- **1 minute**: If still setting up diagram, speed up or skip
- **3 minutes**: Should have horizontal distances calculated
- **5 minutes**: If not near final answer, consider flagging and moving on
- **Pivot Indicator**: If algebra becomes messy with terms > 3 radicals, switch to coordinate approach

Competition Strategy Integration

A. Time Management Strategy

- **Target Time**: 4-5 minutes (appropriate for P15)
- **Maximum Time**: 6 minutes (includes verification)
- **Actual Breakdown**:
  - Reading + diagram: 30 seconds
  - Finding distances: 1.5 minutes
  - Computing areas: 1.5 minutes
  - Final calculation: 1 minute
  - Verification: 30 seconds

B. Problem Prioritization

- **Accessibility**: HIGH - clear geometric setup, standard techniques
- **Difficulty**: MEDIUM-HIGH (P15) - requires careful calculation
- **Point Value**: 6 points (standard AMC12)
- **Strategic Position**: Should attempt after securing P1-14
- **Domain Comfort**: If strong in geometry, prioritize; if weak, consider skipping

C. Risk Management

- **Confidence Threshold**: 90%+ after verification - submit and move on
- **Medium Confidence (70-90%)**: Flag for review if time permits
- **Low Confidence (¡70%)**: Make educated guess between (C) and (D)

D. Abandonment Criteria

Soft Abandon Triggers (Consider switching):

- After 5 minutes without clear path to answer
- Computational errors accumulating (2+ mistakes)
- Algebra becoming unmanageably complex

Hard Abandon Triggers (Immediate switch):

- After 6 minutes total time investment
- Realized fundamental conceptual error in approach
- Emotional frustration affecting clear thinking

Recovery Strategy:

- Mark answer (D) based on answer choice analysis (most complex = likely correct)
- Flag for review if time permits at end
- Move to next problem to maintain momentum

Key Insight

Key Insight: The trapezoid decomposition method completely avoids finding the third side PR of the triangle. By recognizing that:

$$[PP'Q'Q] + [QQ'R'R] = [PQR] + [PP'R'R]$$

we can solve for $[PQR]$ using only simple trapezoid area formulas and the Pythagorean theorem applied to height differences. This transforms a potentially complex Heron's formula calculation into straightforward arithmetic with radicals.

Warning: Common Pitfall

Warning: The most common error is incorrectly applying the Pythagorean theorem. Students often forget that the vertical distance in the trapezoid is the **difference** of the radii (not their sum), and the hypotenuse is the distance between centers (not the sum of radii, though they happen to equal in this case due to external tangency).
Example: For trapezoid $PP'Q'Q$:

- $\times$ WRONG: $P'Q'^2 + 3^2 = 3^2$ (using sum of radii for height)
- $\checkmark$ CORRECT: $P'Q'^2 + 1^2 = 3^2$ (using difference of radii)

Pro Tip

Pro Tip: Answer choice analysis provides a quick confidence check. Choice (D) $\sqrt{6} - \sqrt{2}$ is the most complex expression, suggesting the problem requires multiple radical calculations. When you see your calculation naturally producing $2\sqrt{2}$ and $2\sqrt{6}$, this confirms you're on the right track. In AMC12, more difficult problems (P13-20) often have more complex answer choices.
Additionally, recognize that $\sqrt{6} - \sqrt{2} = \sqrt{2}(\sqrt{3} - 1)$, which could be useful if you need to verify the answer or if the problem asked for the answer in a different form.

Final Answer

$$\boxed{\textbf{(D)}\ \sqrt{6} - \sqrt{2}}$$

Confidence Assessment

- **Confidence Level**: 95% - Clean calculation with exact answer choice match
- **Verification Applied**: Level 4 Standard Verification - dimensional analysis, reasonableness check, algebraic verification all passed
- **Flag for Review**: No - high confidence, all verification checks passed
- **Time Spent**: Estimated 4.5 minutes (within target range)

Alternative Solution Summary

Method 2: Coordinate Geometry with Shoelace Formula

Setup: Place $P' = (0, 0)$ on line l (the x-axis)

- $P = (0, 1)$
- $Q = (2\sqrt{2}, 2)$ (since $P'Q' = 2\sqrt{2}$)
- $R = (2\sqrt{2} + 2\sqrt{6}, 3)$ (since $Q'R' = 2\sqrt{6}$)

Shoelace Formula:

$$\begin{aligned}
\text{Area} &= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\
&= \frac{1}{2}|0(2-3) + 2\sqrt{2}(3-1) + (2\sqrt{2} + 2\sqrt{6})(1-2)| \\
&= \frac{1}{2}|0 + 4\sqrt{2} - 2\sqrt{2} - 2\sqrt{6}| \\
&= \frac{1}{2}|2\sqrt{2} - 2\sqrt{6}| \\
&= \frac{1}{2} \cdot 2|\sqrt{2} - \sqrt{6}| \\
&= |\sqrt{2} - \sqrt{6}| \\
&= \sqrt{6} - \sqrt{2}
\end{aligned}$$

This method produces the same answer with similar computational complexity.

Method 3: Heron's Formula (More Complex)

Step 1: Find the third side PR

$$\begin{aligned}
PR^2 &= (P'R')^2 + (r_R - r_P)^2 \\
&= (2\sqrt{2} + 2\sqrt{6})^2 + (3-1)^2 \\
&= (2\sqrt{2})^2 + 2 \cdot 2\sqrt{2} \cdot 2\sqrt{6} + (2\sqrt{6})^2 + 4 \\
&= 8 + 8\sqrt{12} + 24 + 4 \\
&= 36 + 8\sqrt{12} \\
&= 36 + 16\sqrt{3}
\end{aligned}$$

Therefore: $PR = \sqrt{36 + 16\sqrt{3}} = 2\sqrt{9 + 4\sqrt{3}}$
Step 2: Apply Heron's formula with $a = 3$, $b = 5$, $c = 2\sqrt{9 + 4\sqrt{3}}$
Semi-perimeter: $s = \frac{3+5+2\sqrt{9+4\sqrt{3}}}{2} = 4 + \sqrt{9 + 4\sqrt{3}}$
Area:

$$\begin{aligned}
K &= \sqrt{s(s-a)(s-b)(s-c)} \\
&= \sqrt{(4+a)(4+a-3)(4+a-5)(4+a-2a)} \text{ where } a = \sqrt{9 + 4\sqrt{3}} \\
&= \sqrt{(4+a)(1+a)(a-1)(4-a)} \\
&= \sqrt{[(4+a)(4-a)][(1+a)(a-1)]} \\
&= \sqrt{(16 - a^2)(a^2 - 1)} \\
&= \sqrt{(16 - 9 - 4\sqrt{3})(9 + 4\sqrt{3} - 1)} \\
&= \sqrt{(7 - 4\sqrt{3})(8 + 4\sqrt{3})} \\
&= \sqrt{56 + 28\sqrt{3} - 32\sqrt{3} - 48} \\
&= \sqrt{8 - 4\sqrt{3}}
\end{aligned}$$

Step 3: Simplify nested radical $\sqrt{8 - 4\sqrt{3}}$
Assume $\sqrt{8 - 4\sqrt{3}} = \sqrt{a} - \sqrt{b}$ where $a > b$
Then: $8 - 4\sqrt{3} = a + b - 2\sqrt{ab}$

Comparing: $a + b = 8$ and $2\sqrt{ab} = 4\sqrt{3}$, so $ab = 12$
Solving: $a = 6$, $b = 2$
Therefore: $\sqrt{8 - 4\sqrt{3}} = \sqrt{6} - \sqrt{2}$ ✓
This method is significantly more complex but serves as excellent verification.

Learning Outcomes

Key Techniques Mastered

1. Pythagorean theorem in trapezoid configurations
2. Area decomposition methods for triangles
3. Radical simplification: $\sqrt{8} = 2\sqrt{2}$, $\sqrt{24} = 2\sqrt{6}$
4. Nested radical simplification: $\sqrt{8 - 4\sqrt{3}} = \sqrt{6} - \sqrt{2}$

Strategic Lessons

1. Multiple approaches exist - choose the cleanest calculation
2. Trapezoid decomposition often cleaner than Heron's formula
3. Answer choice complexity provides hints about solution path
4. Verification is crucial for problems with multiple radical terms

Competition Wisdom

1. P15 problems should take 4-6 minutes - budget accordingly
2. Clean geometric diagrams prevent calculation errors
3. Recognize when algebra is getting too messy - switch methods
4. High confidence on mid-band problems justifies moving forward